Class #22 - Basis and Coordinates

1. Def: Let V be a vector space. An indexed set of vectors $\{v_1, v_2, \cdots, v_p\}$ in __________is said to be *linearly independent* if the vector equation

   ____

   has __________the trivial solution $c_1 = 0,\ c_2 = 0, \cdots, c_p = 0.$

2. Theorem: An indexed set $\{v_1, v_2, \cdots, v_p\}$ of two or more vectors, with $v_1 \neq 0$, is linearly dependent if and only if some v_i (i __________) is a linear combination of the preceding vectors, $v_1, \cdots,$ __________.
   Proof:

3. Examples:

   (a) The set $\{1, t, 4 - t\}$ in $\mathbf{P}$ (= The set of all polynomials) is __________.

   (b) The set $\{\sin t, \cos t\}$ in __________is __________.

4. Def: Let H be a subspace of a vector space V. An indexed set of vectors $B = \{b_1, b_2, \cdots, b_p\}$ in __________is a *basis* for H if

   (i) B is a linearly independent set, and

   (ii) Span$B$ = __________.

5. Examples:

(a) Let A be an $n \times n$ invertible matrix. Then the set of the columns of A, $\{a_1, a_2, \cdots, a_n\}$ is a __________of __________.

(b) *The standard basis for* $\mathbf{R^n}$ = __________________________.

(c) *The standard basis for* $\mathbf{P_n}$ = __________________________.

(d) Find a basis for Col$A$ where

$$A = \begin{bmatrix} 1 & 4 & 0 & 2 & -1 \\ 3 & 12 & 1 & 5 & 5 \\ 2 & 8 & 1 & 3 & 2 \\ 5 & 20 & 2 & 8 & 8 \end{bmatrix}$$

It can be shown that $A \sim U = \begin{bmatrix} 1 & 4 & 0 & 2 & 0 \\ 0 & 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$.

Note: The solution set of the equation $Ax = 0$ is same as the solution set of the equation __________. The weights of the equation $Ux = 0$ tells us the relations among the columns of U, and the same weights are the answers to the equation __________, hence the same relations among the columns of A holds.

Observation: Elementary row operations on a matrix do not affect the linear dependence relations among the __________of the matrix.

Basis for Col A = __________________________.

Question: Is Col A = Col U?

6. An important reason for specifying a basis B for a vector space V is to impose a *coordinate system* on V.

7. The Unique Representation Theorem: Let $B = \{b_1, b_2, \cdots, b_n\}$ be a basis for a vector space V. Then for each $x \in V$, there exists a __________set of scalars $c_1, \cdots, c_n$ such that $x = c_1b_1 + \cdots + c_nb_n$.
Proof:

8. Def: Suppose the set $B = \{b_1, b_2, \cdots, b_n\}$ is a basis for V and $x \in V$. The *coordinates of x relative to the basis B)* or the *B-coordinates of x* are the weights $c_1, c_2, \cdots, c_n$ such that ________________________.

   If $c_1, \cdots, c_n$ are the B-coordinates of x, then the vector in __________, $[x]_B = \begin{bmatrix} c_1 \\ \vdots \\ c_n \end{bmatrix}$, is the *coordinate vector of x (relative to B)*, or the *B-coordinate vector of x.*

The mapping $x \mapsto [x]_B$ is the *coordinate mapping (determined by B).*

Fill in the blank: The coordinate mapping from __________to __________is 1) __________, 2) __________, and 3)__________transformation.

9. Example: The set $B = \{1+t, 1+t^2, t+t^2\}$ is a __________ for $\mathbf{P_2}$. (Why?)
   Find $[p(t)]_B$ where $p(t) = 6 + 3t - t^2$.